

Lab 10.6.7 Report: Using Wireshark to Examine HTTP and HTTPS Traffic

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Part 1: HTTP Traffic Capture and Analysis

Objectives

- Capture HTTP network traffic using tcpdump
- Analyze unencrypted HTTP communications
- Identify security vulnerabilities in HTTP protocol

1.2 Network Interface Identification

Command Executed:

ip address

Question: List the interfaces and their IP addresses displayed in the ip address output.

Answer:

- lo (loopback): 127.0.0.1

- enp0s3 (Ethernet): 192.168.1.24

1.3 Packet Capture Process

Command Executed:

sudo tcpdump -i enp0s3 -s 0 -w httpdump.pcap

Command Options Explanation:

- -i enp0s3: Specifies the network interface to monitor
- -s 0: Sets snapshot length to default (262144 bytes) to capture complete packets
- -w httpdump.pcap: Writes captured traffic to specified file

Test Website: <http://www.altoromutual.com/login.jsp> **Test Credentials:**

- Username: Admin

- Password: Admin

1.4 Analysis Results

Question: What two pieces of information are displayed in the HTML Form URL Encoded section?

Answer:

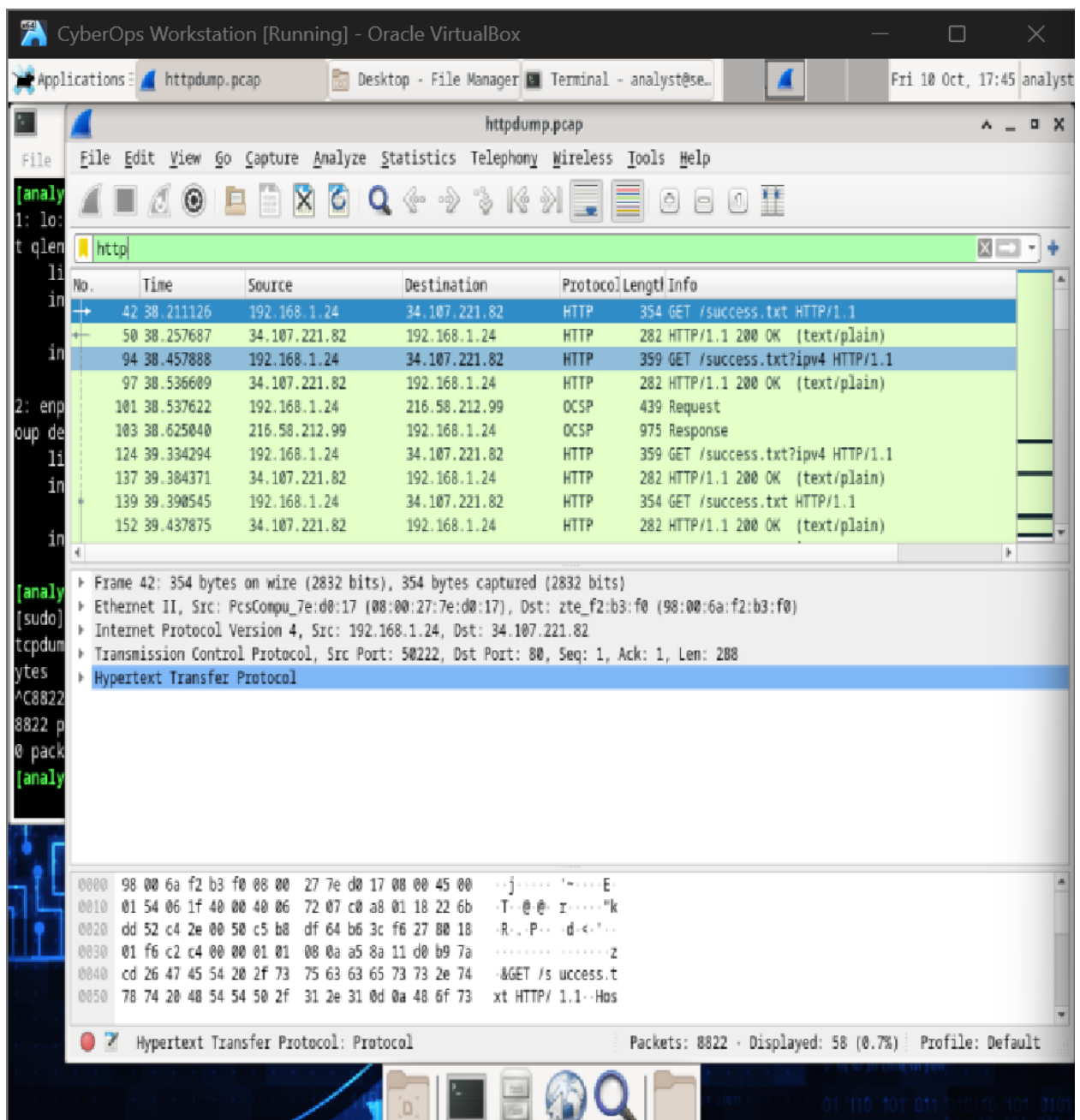
1. uid=Admin (username field)
2. passw=Admin (password field)
3. btnSubnit =Login

Key Observations

Security Vulnerabilities Identified:

1. Login credentials transmitted in plaintext
2. All form data visible to network observers
3. No encryption protecting sensitive information
4. Susceptible to man-in-the-middle attacks
5. Password field clearly readable in Wireshark analysis

Screenshot:



Part 2: HTTPS Traffic Capture and Analysis

Objectives

- Capture HTTPS network traffic
- Compare encrypted vs unencrypted communications
- Understand TLS/SSL encryption mechanisms

Methodology

2.1 HTTPS Packet Capture

Command Executed:

```
sudo tcpdump -i enp0s3 -s 0 -w httpsdump.pcap
```

Test Website: www.netacad.com

Question: What do you notice about the website URL?

Answer:

The website URL begins with "https://" instead of "http://", indicating:

1. Secure connection is established
2. Padlock icon appears in browser address bar
3. SSL/TLS certificate is being used
4. All traffic will be encrypted
5. Green/secure indicator shows verified certificate

2.2 Wireshark Analysis

Filter Applied:

```
tcp.port==443
```

(Port 443 is the standard port for HTTPS traffic)

Question: What has replaced the HTTP section that was in the previous capture file?

Answer:

The HTTP section has been replaced by "TLSv1.2 Record Layer" (or "Secure Sockets Layer").

This layer contains:

- Handshake Protocol information
- Application Data (encrypted)
- Cipher Suite information
- Certificate exchange data

Question: Is the application data in a plaintext or readable format?

Answer:

No, the application data is NOT in plaintext or readable format.

Observations:

- 1. Data appears as encrypted binary content
- 2. No readable text or credentials visible
- 3. Encrypted Application Data section shows hexadecimal values
- 4. Content cannot be decrypted without private keys
- 5. Even form submissions are completely encrypted

Key Observations

Security Features Identified:

- 1. Complete encryption of all transmitted data
- 2. SSL/TLS certificate authentication
- 3. Encrypted handshake process
- 4. No visible credentials or sensitive information
- 5. Protection against eavesdropping and interception

Comparative Analysis: HTTP vs HTTPS

Data Visibility Comparison

Aspect	HTTP	HTTPS
Protocol Layer	Application Layer (plaintext)	Application + TLS/SSL Layer (encrypted)
Password Visibility	Fully visible in Wireshark	Completely encrypted
Data Format	Readable text	Encrypted binary
Port	80	443

Aspect	HTTP	HTTPS
Certificate	None	SSL/TLS Certificate required
Security	No protection	End-to-end encryption
Packet Analysis	Form data readable	Only metadata visible

Security Implications

HTTP Vulnerabilities:

- Credentials captured in plaintext
- Session hijacking possible
- Man-in-the-middle attacks feasible
- Data tampering undetectable
- No authenticity verification

HTTPS Protections:

- Encrypted data transmission
- Certificate-based authentication
- Data integrity verification
- Protection against eavesdropping
- Secure session management

Reflection Questions

Question 1: What are the advantages of using HTTPS instead of HTTP?

Answer:

1. Data Encryption HTTPS encrypts all data transmitted between the client and server using SSL/TLS protocols. This prevents attackers from intercepting and reading sensitive information such as passwords, credit card numbers, and personal data.

2. Data Integrity HTTPS ensures that data cannot be modified or corrupted during transfer. Any tampering attempt will be detected, protecting against man-in-the-middle attacks where attackers might alter transmitted data.

3. Authentication and Trust HTTPS uses digital certificates issued by trusted Certificate Authorities (CAs) to verify the identity of websites. This helps users confirm they are communicating with the legitimate website and not an impostor.

4. Privacy Protection User activities, browsing behavior, and form submissions are protected from network observers, ISPs, and potential eavesdroppers on public networks.

5. Compliance Requirements Many regulations (GDPR, PCI-DSS, HIPAA) require HTTPS for protecting sensitive user data, making it essential for legal compliance.

6. SEO and Performance Benefits Search engines like Google prioritize HTTPS websites in rankings. Modern HTTPS implementations (HTTP/2, HTTP/3) can even provide performance improvements.

7. User Trust and Confidence The padlock icon and security indicators in browsers increase user confidence and reduce bounce rates, especially for e-commerce sites.

Question 2: Are all websites that use HTTPS considered trustworthy?

Answer:

No, not all websites using HTTPS are considered trustworthy.

Important Clarifications:

What HTTPS Does Provide:

1. Encryption of data in transit
2. Verification that you're connected to the domain in the certificate
3. Protection against eavesdropping on the communication

What HTTPS Does NOT Guarantee:

1. The legitimacy of the website owner
2. The trustworthiness of the business or individual
3. Protection against phishing or malicious content
4. The ethical practices of the site operators

Conclusion: HTTPS is a necessary security measure but not sufficient proof of trustworthiness. It should be considered one component of overall website security assessment. Users must remain vigilant and employ multiple methods to verify website legitimacy before sharing sensitive information or conducting transactions.

Practical Applications

Real-World Scenarios

1. Public Wi-Fi Security When using public Wi-Fi networks, HTTPS provides essential protection against:

- Packet sniffing attacks
- Session hijacking
- Credential theft by network observers

2. E-commerce Transactions HTTPS is mandatory for:

- Credit card processing
- Personal information collection
- Account management
- Payment processing

3. Corporate Networks Organizations should:

- Enforce HTTPS for all web applications
- Monitor for HTTP traffic containing sensitive data
- Implement SSL/TLS inspection at network boundaries
- Train employees on secure browsing practices

Tools Learned

tcpdump:

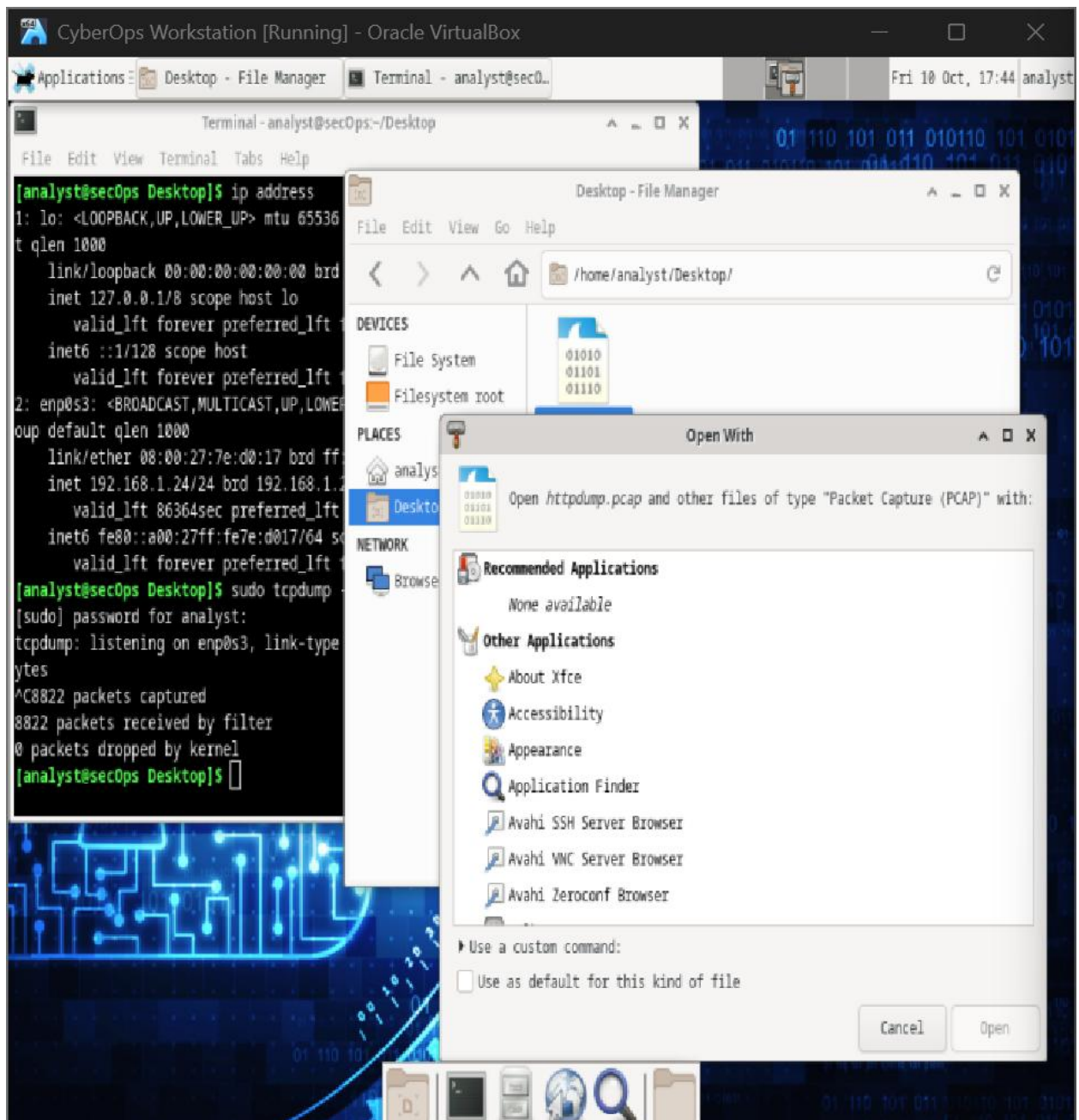
- Command-line packet capture tool
- Lightweight and efficient
- Useful for automated monitoring
- Generates pcap files for later analysis

Wireshark:

- Graphical packet analysis tool

- Detailed protocol dissection
- Filtering and search capabilities
- Essential for network forensics

C. Screenshots



CyberOps Workstation [Running] - Oracle VirtualBox

Applications: httpdump.pcap Desktop - File Manager Terminal - analyst@se... Fri 10 Oct, 17:45 analyst

httpdump.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
42	38.211126	192.168.1.24	34.107.221.82	HTTP	354	GET /success.txt HTTP/1.1
50	38.257687	34.107.221.82	192.168.1.24	HTTP	282	HTTP/1.1 200 OK (text/plain)
94	38.457888	192.168.1.24	34.107.221.82	HTTP	359	GET /success.txt?ipv4 HTTP/1.1
97	38.536609	34.107.221.82	192.168.1.24	HTTP	282	HTTP/1.1 200 OK (text/plain)
101	38.537622	192.168.1.24	216.58.212.99	OCSP	439	Request
103	38.625040	216.58.212.99	192.168.1.24	OCSP	975	Response
124	39.334294	192.168.1.24	34.107.221.82	HTTP	359	GET /success.txt?ipv4 HTTP/1.1
137	39.384371	34.107.221.82	192.168.1.24	HTTP	282	HTTP/1.1 200 OK (text/plain)
139	39.390545	192.168.1.24	34.107.221.82	HTTP	354	GET /success.txt HTTP/1.1
152	39.437875	34.107.221.82	192.168.1.24	HTTP	282	HTTP/1.1 200 OK (text/plain)

Frame 42: 354 bytes on wire (2832 bits), 354 bytes captured (2832 bits)

Ethernet II, Src: PcsCompu_7e:d0:17 (08:00:27:7e:d0:17), Dst: zte_f2:b3:f0 (98:00:6a:f2:b3:f0)

Internet Protocol Version 4, Src: 192.168.1.24, Dst: 34.107.221.82

Transmission Control Protocol, Src Port: 50222, Dst Port: 80, Seq: 1, Ack: 1, Len: 288

Hypertext Transfer Protocol

```
0000 98 00 6a f2 b3 f0 00 00 27 7e d0 17 00 00 45 00  ..j.....E.
0010 01 54 06 1f 40 00 40 06 72 07 c0 a8 01 18 22 6b  -T-@-@-r-...k
0020 dd 52 c4 2e 00 50 c5 b8 df 64 b6 3c f6 27 80 18  -R-.P-..d-<...'
0030 01 f6 c2 c4 00 00 01 01 08 0a a5 8a 11 d0 b9 7a  .....z
0040 cd 26 47 45 54 20 2f 73 75 63 63 65 73 73 2e 74  -&GET /s uccess.t
0050 78 74 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73  xt HTTP/ 1.1..Has
```

Hypertext Transfer Protocol: Protocol Packets: 8822 - Displayed: 58 (0.7%) Profile: Default

